

Yuanshan Zhong

University at Buffalo | Biomedical Engineering

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Research Interests

AI-driven neuroengineering integrating neural signal analysis, representation learning, brain-computer interfaces, computational neurostimulation, and adaptive closed-loop neural systems.

EDUCATION

University at Buffalo, SUNY

Aug 2025 – Present

M.S. in Biomedical Engineering

University of Science and Technology of China (USTC)

Sep 2018 – Jun 2023

B.S. in Computer Science

Special Class for the Gifted Young

RESEARCH EXPERIENCE

Research Assistant

Wenyao Xu Lab, University at Buffalo

Multimodal Neural Signal Analysis for Objective Biomarker Discovery in Auditory Disorders

University at Buffalo | Advisor: Prof. Wenyao Xu | Oct 2025 – Present

- Contributed to the SYNAPSE multimodal neurophysiological platform integrating around-ear EEG and pupillometry for objective assessment of auditory disorders, including tinnitus, hyperacusis, and misophonia.
- Developed EEG analysis pipelines incorporating signal preprocessing, ERP and spectral feature extraction, cross-channel synchronization analysis, and statistical testing; evaluated conventional machine-learning and deep-learning models (EEGNet/EEGNex) for disease-related neural pattern classification and characterization.
- Developing a self-supervised pretraining framework for around-ear EEG by aggregating multiple public datasets and adapting foundation-model representation learning to sparse, non-standard electrode layouts, with the goal of learning transferable neural representations across subjects, datasets, and downstream tasks.
- Applied pretrained EEG foundation models, primarily LaBraM, across electrode channels and auditory tasks to identify latent neural features associated with four auditory conditions, systematically evaluating disease-related representations across channels, temporal windows, and embedding dimensions.
- Investigated what information the foundation model encodes through layer-wise representation analysis, frequency/spatial/temporal feature decomposition, and controlled statistical experiments, tracing how disease-related signals emerge and transform across model layers and distinguishing robust neural structure from potentially confounded representations.

Electrothermal Modeling of High-Density Electrode Grids for Transcutaneous Neurostimulation

University at Buffalo | Advisor: Prof. Filip Stefanovic | Aug 2025 – Present

- Developed coupled electrical–thermal finite-element models in COMSOL Multiphysics to simulate current distribution, Joule heating, and transient temperature responses in multilayer biological tissue during peripheral transcutaneous neurostimulation.
- Systematically evaluated conventional single electrodes and HD-inspired 1–16 electrode grid configurations across 128 combinations of stimulation current, pulse width, and frequency to characterize current-density distribution and tissue heating.
- Characterized how electrode subdivision, active electrode count, and total active area shape current concentration and thermal accumulation, identifying distinct electrothermal regimes and a nonlinear decline in stimulation-side thermal load with increasing electrode count.
- Performed area-matched and count-matched control analyses, mesh-convergence and numerical robustness evaluations, and parameter-space characterization to validate the computational framework and separate geometry-dependent electrothermal effects.

AI-Optimized High-Density Electrode Arrays for Selective Muscle Stimulation

University at Buffalo | Advisor: Prof. Filip Stefanovic | Dec 2025 – Present

- Developing a computational framework combining finite-element modeling, numerical optimization, and machine learning to optimize current steering across high-density surface electrode arrays for selective activation of targeted muscle regions.
- Constructing an 8×8 independently controlled electrode-array FEM model with anisotropic muscle conductivity, using precomputed basis fields and linear superposition to efficiently evaluate arbitrary 64-channel current distributions.
- Building a superposition-based optimization pipeline to generate optimal electrode-current distributions across target locations, depths, and muscle-fiber orientations, and training inverse ML surrogate models for rapid prediction of stimulation configurations.
- Extending the framework toward anatomically informed stimulation models by incorporating heterogeneous tissue properties, realistic muscle geometry and fiber architecture, with the long-term goal of adapting a standardized optimization framework to specific body regions.

Hybrid Learning for Robust Neural Control in Brain–Computer Interfaces

Independent Research | Apr 2025 – Present

- Developed an EEG-driven closed-loop neural control framework for 2D cursor navigation using BCI Competition IV-2a, integrating neural decoding with sequential decision-making in an interactive control environment.
- Developed a hybrid supervised–reinforcement learning architecture that compresses EEG features into structured low-dimensional control states and jointly integrates neural representations, decoder outputs, and task geometry for policy learning.
- Evaluated the framework under strict cross-subject ID/OOD splits and EEG perturbations, demonstrating improved convergence, training stability, cross-subject generalization, and robustness of compact neural control representations.

- Extending the framework toward adaptive closed-loop neural control with continuously updated EEG representations, time-varying task states and goals, and feedback-driven policy adaptation in dynamic environments.

Event–Frame Hybrid Neural Network for Occlusion Removal

University of Science and Technology of China | Advisor: Prof. Xueyang Fu | Dec 2022 – Jul 2023

- Developed a multimodal framework integrating event streams and image frames.
- Designed a hybrid neural network combining spiking neural networks (SNN) and convolutional networks.
- Developed cross-modal feature fusion to improve reconstruction under occlusion conditions.
- Constructed a dataset with 900+ samples and evaluated reconstruction performance using PSNR and SSIM.

Knowledge-Graph-Based Recommendation with Graph Representation Learning

Research Project | University of Science and Technology of China | 2022

- Studied the integration of knowledge graphs and collaborative filtering for explainable movie recommendation.
- Built a knowledge graph from Freebase entities linked to Douban movies and extracted multi-hop relational structures for representation learning.
- Implemented knowledge graph embedding models (TransE, TransR) and integrated them with matrix factorization to enhance item embeddings.
- Evaluated recommendation performance using ranking metrics (Recall@5/10, NDCG@5/10).

Programming Language Extension and Compiler Implementation

Research Project | University of Science and Technology of China | 2021

- Implemented a YACC-based parser to resolve ambiguity in relational expression grammars and enable correct evaluation of chained comparison expressions.
- Extended the educational PL/0 compiler with multi-dimensional array support, including symbol-table representation and runtime memory layout computation.
- Developed compiler backend mechanisms for array indexing, assignment, and control flow (for-loops and logical operators) through stack-machine instruction generation.

PUBLICATIONS & MANUSCRIPTS

Zhong, Y., & Stefanovic, F. (2026). Comparative electrothermal analysis of single and small-electrode grid configurations for HD-inspired peripheral transcutaneous neurostimulation. *Frontiers in Bioengineering and Biotechnology*. [doi:10.3389/fbioe.2026.1904885](https://doi.org/10.3389/fbioe.2026.1904885)

Zhong, Y.*, Das, A.*, Rivera Rosario, E., Sun, W., & Xu, W. (2026). From decoding performance to interpretation in EEG foundation-model representations. **Manuscript in preparation (submission planned for September 2026).**

*Co-first authors.

Zhong, Y., & Stefanovic, F. (2026). AI-optimized high-density electrode arrays for selective muscle stimulation. **Manuscript in preparation.**

TECHNICAL SKILLS

Neuroengineering

EEG signal processing, neural representation learning & analysis, EEG foundation models, ERP analysis, time-frequency analysis

Machine Learning

PyTorch, deep learning, self-supervised learning, reinforcement learning, model interpretability

Modeling & Simulation

COMSOL Multiphysics, finite-element modeling, electrothermal modeling, bioelectric field simulation, numerical optimization

Programming

Python, MATLAB

Scientific Computing & Data Analysis

NumPy, SciPy, scikit-learn, MNE-Python

HONORS & AWARDS

National Scholarship (Third Prize), University of Science and Technology of China, 2018

Third Prize, Provincial Level Physics High School Competition (China)

First Prize, City-Level Mathematics High School Competition (China)